

Program/s: MBA Tech/B.Tech Data Science

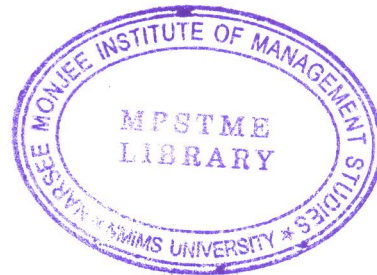
Year: II Semester: IVStream/s ; Data Science

Subject: Statistical Structures in Data and Inference

Time: 3 hrs (~~10:00am~~ to 1:00 pm)Date: 28 / 06 / 2023No. of Pages: 3Marks: 100**Re-Examination (2021-22/2022-23)**

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.



Q1		Answer briefly:	[20]																		
CO-1 ; SO- ;1 BL-2	a.	What is the difference between R square and adjusted R square?	[5]																		
CO-1 ; SO- ;1 BL-2	b.	How to interpret alpha and beta in the results of a logistic regression model?	[5]																		
CO-2 ; SO- ;1 BL-4	c.	What's the Difference Between a T-Test and Z-Test?	[5]																		
CO-1 ; SO- ;3 BL-2	d.	How is Hypothesis Testing using in Linear Regression?	[5]																		
Q2			[20]																		
CO-3;SO-2; BL-4	A	<i>Python code:</i> <pre>import pandas as pd import statsmodels.formula.api as smf df1= pd.DataFrame({'x':[89,43,36,36,95,10,66,34,38,20,26,29,48,64,6,5,36,66,72,40], 'y':[21,46,3,35,67,95,53,72,58,10,26,34,90,33,38,20,56,2,47,15]})</pre> Check whether x has a non-linear relationship with Y or not. Produce python code and analyze the output.	[10]																		
CO-2; ;2,12 BL-4	B	The times required by three workers to perform an assembly-line task were recorded on five randomly selected occasions. Here are the times, to the nearest minute. <table><tr><td>Hari</td><td>Ram</td><td>Jay</td></tr><tr><td>8</td><td>8</td><td>10</td></tr><tr><td>10</td><td>9</td><td>9</td></tr><tr><td>9</td><td>9</td><td>10</td></tr><tr><td>11</td><td>8</td><td>11</td></tr><tr><td>10</td><td>10</td><td>9</td></tr></table>	Hari	Ram	Jay	8	8	10	10	9	9	9	9	10	11	8	11	10	10	9	[10]
Hari	Ram	Jay																			
8	8	10																			
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		Preliminary data analyses indicate that the independent samples come from normal populations with equal standard deviations. At the 5% significance level, does there appear to be a difference in mean assembly-line task?	
Q3			[20]
CO-3; SO-;2,5 BL-3	A.I	Before and after as objects containing the scores of python training in a company. Find out whether the training program is effective or not. <i>Python code:</i> <pre>from scipy import stats</pre> after = [12.2, 14.6, 13.4, 11.2, 12.7, 10.4, 15.8, 13.9, 9.5, 14.2] before = [13.5, 15.2, 13.6, 12.8, 13.7, 11.3, 16.5, 13.4, 8.7, 14.6] Produce python code and analyze the output	[5]
CO-3; SO-;2,5 BL-3	A.I I	Below data hold plant heights from each sample <i>Python code:</i> <pre>G1 <- [8, 8, 9, 9, 9, 11, 12, 13, 13, 14, 15, 19]</pre> <pre>G2<- [11, 12, 13, 13, 14, 14, 14, 15, 16, 18, 18, 19]</pre> Find out whether the mean height of both groups are same or not. Produce python code and analyze the output	[5]
CO-1; SO-;2,5 BL-4	B	Reena believes that 50% of first-time brides in the India are younger than their grooms. She performs a hypothesis test to determine if the percentage is the same or different from 50%. Reena samples 100 first-time brides and 53 reply that they are younger than their grooms. Analyze whether what Reena believes is true or false at 1% level of significance.	[10]
Q4			[20]
CO-3; SO-3; BL-4	A	The Data "salary" contains 2008-09 nine-month academic salary for Assistant Professors, Associate Professors and Professors in a college in the U.S. First, build a regression model considering salary as dependent variable and others as independent variable. Secondly build a regression model considering qualitative variable as only independent variables and interpret the results.	[10]
CO-3; SO-;3 BL-3	B	Produce python code to calculate mean and standard deviation through Maximum likelihood estimation in the below data set refer as df. <i>Python code:</i> <pre>import numpy as np</pre> <pre>from scipy.optimize import minimize</pre> <pre>from scipy.stats import norm</pre> <pre>np.random.seed(123)</pre> <pre>N=500</pre> <pre>mu =5</pre> <pre>sigma =9</pre> <pre>df = np.random.normal(loc = mu, scale=sigma,size=(N,))</pre>	[10]
Q5			[20]

CO-1; SO-;1 BL-2	A	Write down the steps of testing of Hypothesis with an examples.	[10]																						
CO-1; SO-;2 BL-3	B	Given the time between the arrival of two devotees at a temple is exponentially distributed with rate λ , we have observed the time between the arrivals as (1.34, 3.67, 5.21, 4.15). For which value of λ is the probability of the observed sample the largest?	[10]																						
Q6			[20]																						
CO-1; SO-;1 BL-2	A	Bank of India, Mumbai branch manager shows that the median number of savings account customers per day is 64. A clerk from the same branch claims that it was more than 64. The clerk found the number of savings accounts customers per day data for 10 random days. Can we reject the branch manager's claim at a 0.05 significance level? (*Use sign test) <table border="1"><tr><td>Day</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><td>Customers</td><td>60</td><td>66</td><td>65</td><td>70</td><td>68</td><td>72</td><td>46</td><td>76</td><td>77</td><td>75</td></tr></table>	Day	1	2	3	4	5	6	7	8	9	10	Customers	60	66	65	70	68	72	46	76	77	75	[10]
Day	1	2	3	4	5	6	7	8	9	10															
Customers	60	66	65	70	68	72	46	76	77	75															
CO-1; SO-2; BL-1	B	Briefly explain multiple regression. Write a program(Python) on advertisement dataset to calculate regression coefficients to predict product sales based on advertising budgets on TV, radio and in newspapers.	[10]																						
Q7			[20]																						
CO-1; SO-2; BL-1	A	A baseball coach wants to know if his team is representative of other teams in scoring runs. Nationally, the average number of runs scored by a team in a game is 5.7. He chooses five games at random in which his team scored 5, 9, 4, 11, and 8 runs. Is it likely that his team's scores could have come from the national distribution? Assume the level of significance as 0.05.	[10]																						
CO-3; SO-2; BL-4	B	Briefly explain logistic regression. Build a logistic regression model on the above data set. Produce python code, analyze the output and confusion matrix. <i>Python code:</i> <pre>from sklearn import linear model from sklearn.metrics import confusion_matrix,classification_report df3=pd.DataFrame({'Age':[25,32,24,27,33,35,48,50,55,45,60,65], 'Salary':[15000,74000,79000,40000,72000,65000,20000,40000,60000,52000,40000, 45000], 'y':[0,0,0,0,0,0,1,1,1,1,1,1]})</pre>	[10]																						